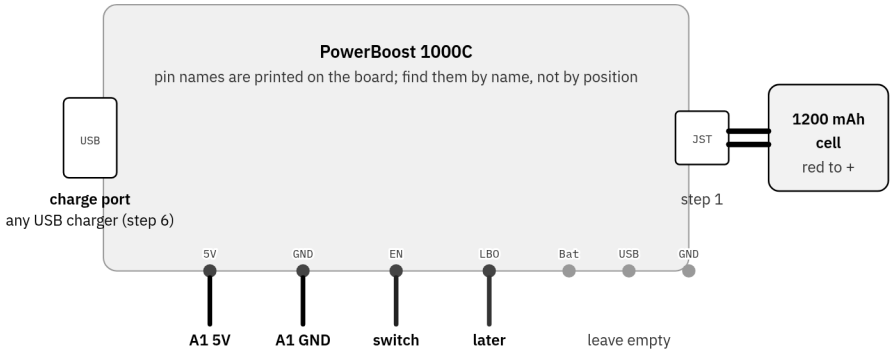
POCKET AI ASSISTANT · BATTERY · THE REAL TEST

A1 on the PowerBoost: the real test

Eight steps, one rule. Needs the Adafruit PowerBoost 1000C, the 1200 mAh cell, two short wires, and the server running on the computer. The result shows up in the server log over Wi-Fi, so A1 needs no USB cable at all.

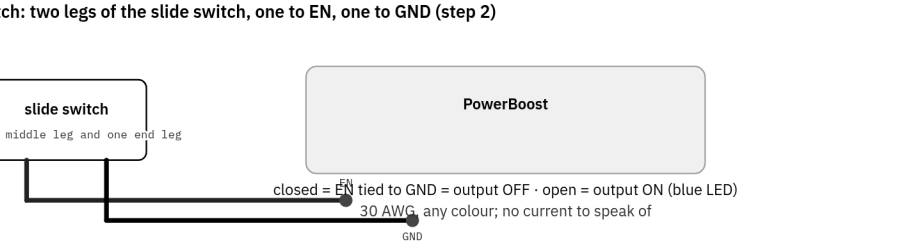
> **THE ONE RULE**
> A1's USB cable and the PowerBoost output are never connected at the same time. Switch the PowerBoost off before plugging A1 in; unplug A1 before switching on.

- [ ] **Plug the cell's JST into the PowerBoost's JST socket, red toward the + mark on the board.**
  CHECK · Blue power LED comes on.



Five places on the PowerBoost. The cell's JST, the charge port, and three pins: 5V, GND, EN. LBO is for later. Pin positions on your board may differ from this sketch; the names printed beside them do not.

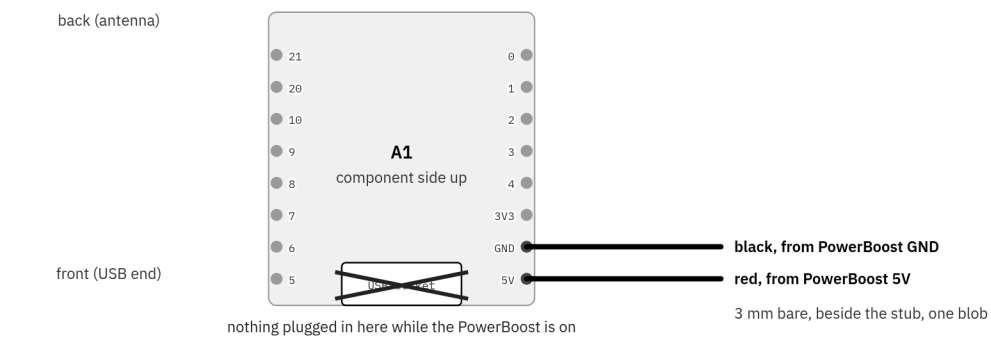
- [ ] **Wire the switch: one leg to `EN`, the other to `GND` on the PowerBoost (30 AWG). Slide it to the position where the blue LED goes *off*. That is OFF.**
  CHECK · Blue LED off.



The switch is just a way to tie EN to GND. Closed: output off. Open: output on. Which slider position is which, the blue LED tells you.

- [ ] **With the switch OFF: PowerBoost `5V` → A1's 5V stub (red 26 AWG), PowerBoost `GND` → A1's GND stub (black). A1's own USB: nothing plugged in.**

  CHECK · Two wires, two joints, A1's USB empty.



At A1: the two stubs nearest the USB end of the right row. 5V is the very first, GND the second. The red wire on 5V, the black on GND; the 5 V rod and the GND rod of the harness share those same stubs.

- [ ] **Server running on the computer (`http://192.168.1.171:8790/health` says `ok`). Switch ON.**

  CHECK · Blue LED on; eyes open on the screen; the server log prints `hello accepted` within ten seconds and `/health` lists `pocket-a1` .

- [ ] **Hold the button, ask Pip its name, release.**

  CHECK · Transcript in the log; the face reacts (and speaks, if the amp build is loaded).

- [ ] **Plug a USB charger into the PowerBoost's own port while Pip is on.**

  CHECK · Eyes stay on; yellow LED = charging while running. That is load sharing.

- [ ] **Unplug the charger.**

  CHECK · Eyes stay on; blue LED only. Pip is on the cell.

- [ ] **Switch OFF.**

  CHECK · Eyes off, blue LED off. Done: boost, switch, load sharing and cell power all proven.

If step 4 gives no eyes: meter `5V` to `GND` on the PowerBoost with the switch on (5.1–5.3 V); then the two wires at A1's stubs. If the log shows `got ip` but no `hello` , it is the server, not the power.

PowerBoost 1000C pins and LEDs from Adafruit's product page; the firmware log lines from `character/firmware` as proven on the board on 10 October. Not yet run on the PowerBoost.